

# Cardinal theory of utility

Explain cardinal theory or approach of consumer behaviour.

Cardinal means numbers and the proponents of this approach advocate that the utility can be measured in numbers. Therefore the economists analyze consumer's behaviour through measuring utility with hypothetical numbers. This approach consists of following two laws or principles:

- 1- Law of DMU
- 2- Law of EMU

Date: \_\_\_\_\_

Day: \_\_\_\_\_

# Economics

## law of DMU

Def:

According to law of DMU

"If other things don't change, then with the continuous use of a commodity utility of extra or next unit i.e. MU diminishes."

Math Explanation:

If consumer consumes more units of a good, its MU decreases.

$$MU = f(Q)$$

$$MU \uparrow \quad \& \quad Q \downarrow$$

$$MU \downarrow \quad \& \quad Q \uparrow$$

MU depends upon Quantity of a commodity being



consumed.

## Explanation

Table:

| units of glass | MU | TU |
|----------------|----|----|
| 1              | 20 | 20 |
| 2              | 15 | 35 |
| 3              | 10 | 45 |
| 4              | 5  | 50 |
| 5              | 0  | 50 |
| 6              | -5 | 45 |

Explanation:

i) In table we show units in 1<sup>st</sup> column, MU in 2<sup>nd</sup> column and TU in 3<sup>rd</sup> column.

ii) If consumer consumes unit of glass continuously MU decreases. A point will come where MU becomes zero. This is called the point of satisfaction.

iii)  $MU > 0$  before S.P is +ve

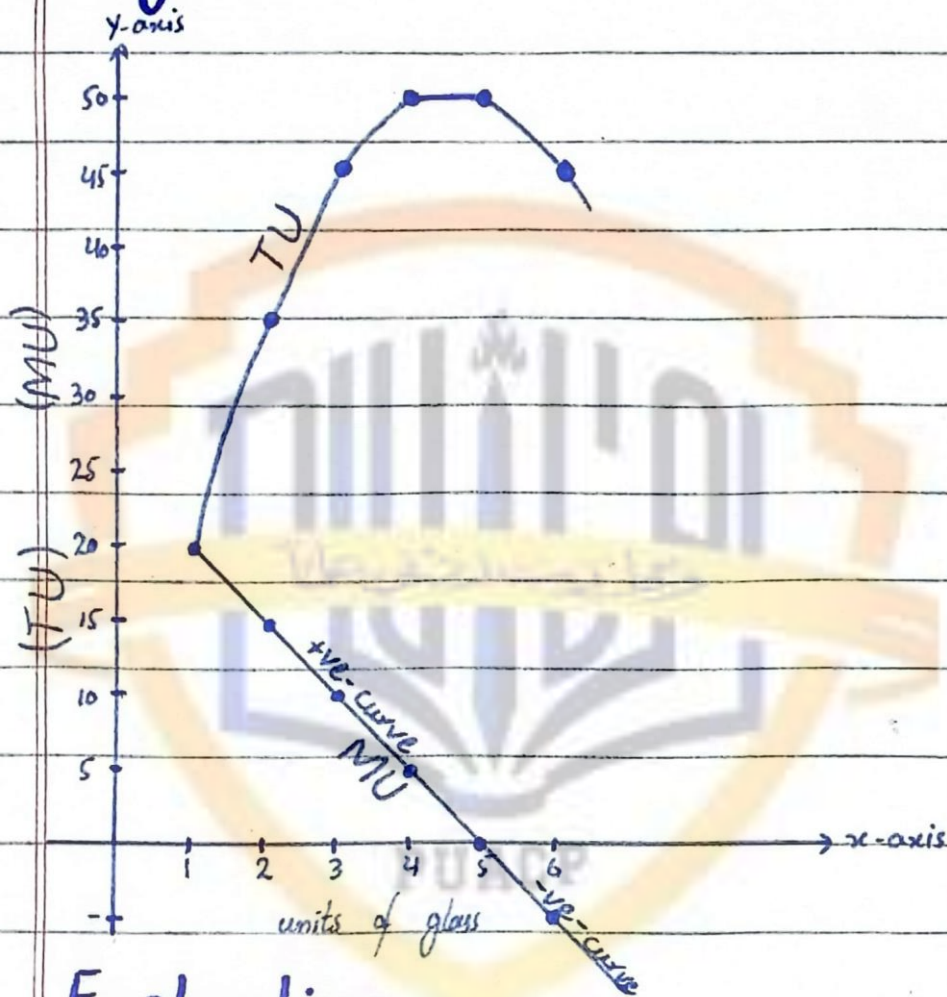
Date: \_\_\_\_\_

Day: \_\_\_\_\_

and after is -ve.

- iv) At point 6 TU is highest and MU is lowest that is 0.

**Diagram:**



**Explanation:-**

- i) In diagram at x-axis we shown units of glass and at y-axis we shown MU and TU.

- ii) MU curve is drawn using units of glass and MU. TU curve is drawn using unit of glass



and TU.

iii) The upper part of MU-curve is called +ve curve and lower part is called -ve curve.

iv) From (ta) diagram it is shown that when unit of glass are increased the utility is decreased.

v) A point will come where MU is zero this point is called point of satisfaction.

vi) When MU is 0, TU is highest that is so.

### Assumptions:-

Different assumptions of DMU are as follows:-

i) Continuous use:-

Law of DMU is applicable only in case when consumer consumes unit of commodity continuously.

Date: \_\_\_\_\_

Day: \_\_\_\_\_

ii) **Homogenous Units:-**

Law of DMU

is applicable only in case when all units of commodity are homogenous. like size and quality.

iii) **Consumer's income:-**

Law of

DMU is applicable only in case when consumer's income remains constant.

iv) **Consumer's behaviour:-**

Law of

DMU is applicable only in case when consumer's behaviour not change.



Date: \_\_\_\_\_

Day: \_\_\_\_\_

## Law of EMU

### Definition:-

According to law of EMU:

"If other things remain constant and consumer spend his money on diff goods in such a way that MU of all goods being purchased become equal per unit of money then TU become maximum and this situation is called consumer equilibrium."

### Math Explanation:-

Mathematically  
law of EMU can be written as:

$$C.E = \left( \frac{MU^A}{P^A} = \frac{MU^B}{P^B} = \dots = \frac{MU^N}{P^N} \right)$$

Date: \_\_\_\_\_

Day: \_\_\_\_\_

## Explanation

We explain the law of EMU with help of table and diagram:

Table:

| units  | $MU^x$ | $MU^y$ |
|--------|--------|--------|
| 1      | 14     | 12     |
| 2      | 12     | 10     |
| 3      | 10     | 8      |
| 4      | 8      | 6      |
| 5      | 6      | 4      |
| Total: | 50     | 40     |

Explanation:-

In table we shown units in 1<sup>st</sup> column,  $MU$  of  $x$  in 2<sup>nd</sup> column and  $MU$  of  $y$  in 3<sup>rd</sup> column. By this table it is prove that when the  $MU^x = MU^y$  the total utility is maximum and this situation is called consumer equilibrium.

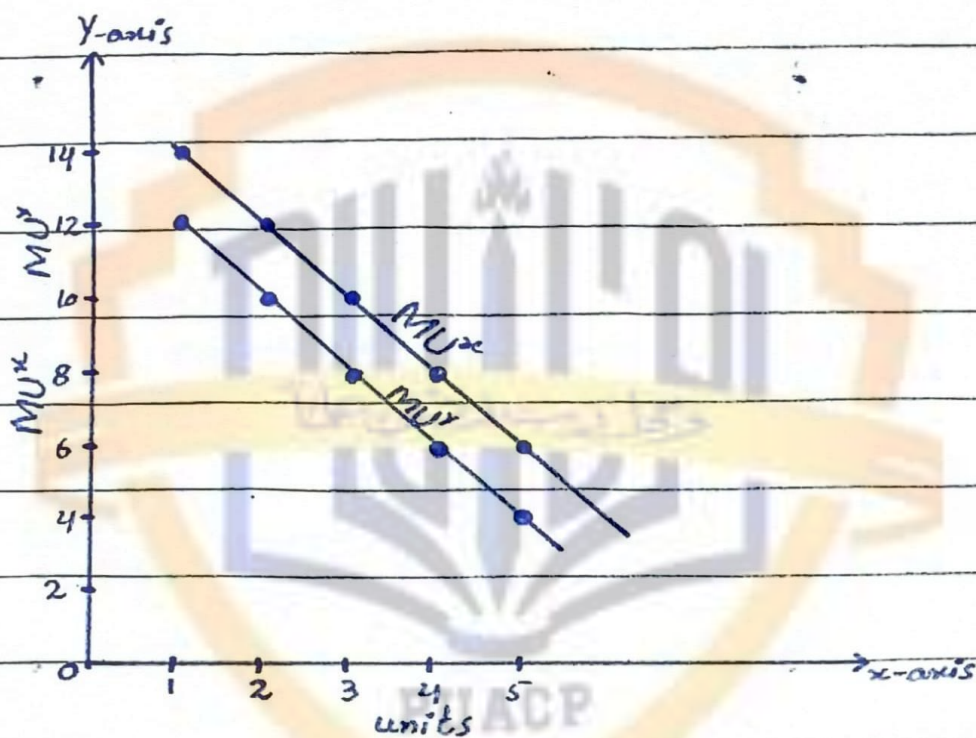


Date: \_\_\_\_\_

Day: \_\_\_\_\_

If  $MU^x \neq MU^y$  then TU is not maximum.

**Diagram:-**



**Explanation:-**

In diagram on x-axis we shown units and on y-axis we shown  $MU^x$  and  $MU^y$ . We draw  $MU^x$  curve using units and  $MU^x$ , and  $MU^y$  curve using units and  $MU^y$ . If the  $MU^x = MU^y$ , then TU is maximum.

Date: \_\_\_\_\_

Day: \_\_\_\_\_

and if  $MU^x \neq MU^y$ , then TU is not maximum.

### Assumptions:-

Different assumptions for the EMU are as follows:-

#### Rational consumer:-

Law of EMU is applicable in a case when consumer is rational.

#### Units of money:-

Law of EMU is applicable in a case when a consumer has some amount of money.

#### Commodities:-

Law of EMU is applicable in a case when consumer wants to purchase different commodities.

#### Price:-

Law of EMU is applicable in a case when the price of all commodities is equal.